

Arterial Blood Gas Analysis: Fundamentals, Interpretation, and Clinical Utility

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Introduction

Blood gas analysis is a commonly used diagnostic tool that evaluates the partial pressures of gases and the acid-base content of a blood sample. Understanding and using blood gas analysis enables clinicians to interpret respiratory, circulatory, and metabolic disorders.[1] Although blood gas analysis is most frequently ordered by clinicians in emergency medicine, critical care, anesthesiology, and pulmonology, it has broad applications across diverse clinical settings. Common indications include acute respiratory distress syndrome (ARDS), severe sepsis, septic shock, hypovolemic shock, diabetic ketoacidosis (DKA), renal tubular acidosis, acute respiratory failure, heart failure, cardiac arrest, asthma, and inborn errors of metabolism.[2]

Blood gas analysis can be performed on blood obtained from any part of the circulatory system, including arteries, veins, or capillaries. An arterial blood gas (ABG) specifically analyzes blood obtained from an artery and measures the arterial partial pressures of oxygen (PaO_2) and carbon dioxide (PaCO_2).[3] PaO_2 provides information on the patient's oxygenation status, whereas PaCO_2 provides information on the patient's ventilation status and helps identify disorders such as chronic or acute respiratory failure. PaCO_2 is affected by hyperventilation (rapid or deep breathing), hypoventilation (slow or shallow breathing), and acid-base status.[2] Although oxygenation and ventilation can be assessed noninvasively using pulse oximetry and end-tidal carbon dioxide (CO_2) monitoring, respectively, ABG analysis remains the gold standard.[4]

When assessing acid-base balance, most ABG analyzers directly measure pH and PaCO_2 .[3] A derivative of the Henderson-Hasselbalch equation is used to calculate serum bicarbonate (HCO_3) and the base deficit or excess. The accuracy of this calculation has been disputed depending on the study, machine, or calibration used, and it must be interpreted appropriately in accordance with institutional standards.[5] Because blood gas analysis calculates HCO_3 from measured pH and PaCO_2 values, whereas serum chemistry analysis directly measures total CO_2 content, small discrepancies between the reported and measured values are expected.[5][6] The difference typically amounts to around 1.2 mmol/L. However, a more considerable difference may be observed in specific clinical scenarios, especially in critically ill patients.[7]

Pathophysiology

ABG analysis provides insight into pulmonary gas exchange, ventilation, and acid-base homeostasis by measuring arterial pH and the partial pressures of oxygen and carbon dioxide.[1] Comparison of pH, PaCO_2 , and HCO_3 values helps identify primary respiratory and metabolic acid-base disturbances as well as compensatory physiologic responses. Tissue hypoxia, impaired ventilation, ventilation-perfusion (V/Q) mismatch, and abnormal cellular metabolism can all alter ABG parameters. The alveolar-arterial (A-a) oxygen gradient is a useful measure of pulmonary gas exchange and may be elevated in conditions associated with V/Q mismatch, diffusion impairment, or intrapulmonary shunting.[8]

Specimen Requirements and Procedure

Whole blood is the required specimen for ABG analysis. The specimen is obtained through an arterial puncture or acquired from an indwelling arterial catheter. Please see StatPearls' companion resource, "[Arterial Lines](#)," for further information. Once obtained, the arterial blood sample should be placed on ice and analyzed as soon as possible to reduce the possibility of erroneous results.[9]

Measurement of PaO₂ is the primary reason for obtaining an arterial specimen, as differences in measured blood gas values between arterial and venous blood are most pronounced for the partial pressure of oxygen (PO₂).

[10] Typically, PO₂ is approximately 60 mm Hg lower in venous blood after the oxygen is released in the capillaries. In contrast, the partial pressure of carbon dioxide (PCO₂) is 2 to 8 mm Hg higher, and pH is only 0.02 to 0.05 lower in a venous sample.[11] Proper specimen collection is paramount in obtaining accurate results. Placement of indwelling catheters with heparin locks for short- and long-term intravenous therapies is typical. Failure to properly flush the lock can have unpredictable effects on measured quantities and is often indicated by unexpected, nonphysiologic results.[12]

Arterial or venous specimens should be collected anaerobically in 1- to 3-mL sterile syringes using lyophilized heparin as the anticoagulant. Evacuated lithium heparin collection tubes used for plasma sampling are unsuitable for blood gas analysis because residual air and ongoing gas exchange may affect measured PO₂ values.[13] Syringes containing lyophilized heparin are preferred over those containing liquid heparin, as liquid heparin has atmospheric PO₂ and PCO₂ values that dilute the sample.[14] When the syringe is incompletely filled, the effect is most pronounced, as a higher ratio of liquid heparin to blood can have a more marked effect on measured PCO₂ and the parameters derived from it. Variations in syringe manufacturing can also create significant preanalytical effects on the sample.[15]

Anaerobic collection techniques minimize exposure of the blood sample to atmospheric air.[16] The PCO₂ of atmospheric air is approximately 0.25 mm Hg, which is much lower than that of blood (40 mm Hg). Thus, the total CO₂ content and PCO₂ measurement of a blood sample exposed to air decrease, and blood pH, as a function of PCO₂, subsequently rises. The PO₂ of atmospheric air (155 mm Hg) is approximately 60 mm Hg higher than that of arterial blood and approximately 100 mm Hg higher than venous blood. Hence, blood from a patient breathing room air absorbs oxygen when exposed to atmospheric air, whereas blood with a PO₂ exceeding 150 mm Hg, as observed in patients undergoing oxygen therapy, releases oxygen.[17]

Air exposure can occur simply from the air in the needle and the dead space in the syringe hub. The error is minimal if the resulting bubble is ejected immediately after drawing by holding the syringe tip up and ejecting a small drop of blood.[18] The potential effect of small bubbles on blood gas results was demonstrated in one study in which a 100 mL bubble of room air was added to 10 blood samples (2 mL each) with PO₂ values ranging from 25 to 40 mm Hg. PO₂ increased an average of 4 mm Hg in these samples in only 2 minutes, and PCO₂ decreased by 4 mm Hg. Before analysis, the sample should be mixed by vigorously rolling the syringe between the technician's palms to ensure homogeneity.[19] Arterialized capillary blood is sometimes an acceptable alternative to arterial blood when an arterial cannula is unavailable or repeated arterial punctures must be avoided.[20]

Automated blood gas analyzers are commonly used to analyze blood gas samples, and results are obtained within 10 to 15 minutes. These automated analyzers, directly and indirectly, measure specific components of the arterial blood gas sample.[1] In the case of an ABG, the analyzed components include the following:

- Measured:
 - pH = acid-base balance of the blood
 - PaO₂ = partial pressure of oxygen in arterial blood

- PaCO_2 = partial pressure of carbon dioxide in arterial blood
- Calculated:
 - HCO_3^- = concentration of bicarbonate in arterial blood
 - Base excess/deficit = relative excess or deficit of base in arterial blood
 - SaO_2 = arterial oxygen saturation (unless a co-oximetry is obtained, in which case SaO_2 is measured)

The radial artery is the preferred site for ABG because it is superficial and easily palpable over the radial styloid process. The femoral artery is the next most common site. Before an ABG is drawn from either upper extremity, a modified Allen test is necessary to confirm adequate collateral blood flow. The test is performed on the unilateral upper extremity chosen for the procedure (see **Image**. Modified Allen Test). The patient is asked to flex the selected upper extremity at the elbow and clench a raised fist for 30 seconds. The clinician applies pressure over the ulnar and radial arteries to occlude the blood flow. After 5 seconds, the patient may unclench the raised fist. At this point, the palm appears pale, white, or blanched. Then, pressure over the ulnar artery is released while the radial artery compression is maintained. In 10 to 15 seconds, the palm should return to its original color, indicating adequate ulnar collateral blood flow. If the palm does not return to its original color, the result is considered abnormal, indicating that puncturing the radial artery is unsafe. Similarly, the radial collateral blood flow is assessed by maintaining ulnar artery pressure and releasing the radial artery pressure.[21] As alternatives to the modified Allen test, pulse oximetry and duplex ultrasound can also be used to assess collateral flow.

Testing Procedures

Operating a traditional blood gas instrument begins with the operator presenting a blood specimen at the sample probe. The sample is taken through the probe by a peristaltic pump, which loads the chamber with a specific amount of the sample. The sample then resides in the chamber long enough to allow thermal equilibration and completion of the measurements. On completion, the pump pushes the sample to waste.[22] Because the instrument's electrodes are not stable for very long, frequent calibration of pH, PCO_2 , and PO_2 is required.[23] Most instruments include a barometer, ensuring that barometric pressure is always known to the microprocessor during calibration. Other instruments perform point-of-care or bedside testing. Almost all manufacturers now produce small, portable, stand-alone, easy-to-operate instruments designed for satellite laboratory use; several hand-held devices that use disposable electrodes are also available.[24]

The sophistication of contemporary equipment and the availability of high-quality calibrator materials have improved the reliability and accuracy of blood gas and pH determination, particularly when combined with meticulous maintenance, adherence to manufacturer-recommended procedures, appropriate equipment control, and proper specimen collection and handling.[22] Software for the instrument's microprocessor often includes display warnings and diagnostic routines that alert the operator and assist with troubleshooting. The manufacturer's recommended maintenance schedule should be considered a minimum guideline, with experience indicating the appropriate maintenance frequency.[25]

Cleanliness of the sample chamber and path is essential. Automatic flushing to cleanse the sample chamber and path after each blood sample measurement is a feature of most instruments that do not use disposable electrodes. Despite proper flushing, however, complete or partial clogging of the chamber or path may occur.[1] Fibrin threads and small clots may be present in the specimen or may form while the sample resides in the warm chamber. If left in place, the debris can affect subsequent measurements or calibrations by interfering with contact between blood, buffers, or gases and electrode membranes.[18] Visibility of the path through the heat sink helps detect clogs, dirt, and bubbles. Bubbles that fail to rinse out can be problematic if they settle on an electrode.[26]

Results, Reporting, and Critical Findings

Normal reference ranges for ABG parameters are listed below, although values may vary slightly among laboratories and across age groups, from neonates to older adults:

- pH (7.35-7.45)
- PaO₂ (75-100 mm Hg)
- PaCO₂ (35-45 mm Hg)
- HCO₃ (22-26 mEq/L)
- Base excess/deficit (−4 to +2)
- SaO₂ (95%–100%) [27][28]

Arterial blood gas interpretation should be approached systematically to assess the severity and chronicity of any abnormalities and determine whether the primary disorder is metabolic or respiratory in origin.[29] Several articles have described simplistic methods for interpreting ABG results. However, the Romanski method is among the most straightforward and is suitable for clinicians at all levels of training. This method helps determine the presence of an acid-base disorder, its primary cause, and whether compensation is present.[30]

The first step of the Romanski method is to assess the pH for the presence of acidemia (pH <7.35) or alkalemia (pH >7.45). If the pH is within the normal range (7.35-7.45), a pH of 7.40 should be used as the cutoff; in other words, clinicians should categorize a pH of 7.37 as acidosis and 7.42 as alkalemia. Next, the respiratory and metabolic components of the ABG results should be evaluated: the PaCO₂ and HCO₃, respectively. The PaCO₂ indicates whether the acidosis or alkalemia is primarily from a respiratory or metabolic acidosis/alkalosis. PaCO₂ greater than 40 mm Hg with a pH less than 7.4 indicates a respiratory acidosis, whereas PaCO₂ less than 40 mm Hg and pH greater than 7.4 indicates a respiratory alkalosis. Respiratory alkalosis is often the result of hyperventilation from either anxiety or compensation for a metabolic acidosis. The next step is to determine whether compensation has occurred by identifying (PaCO₂ or HCO₃) is inconsistent with the pH. Finally, PaO₂ should be assessed for any abnormalities in oxygenation.[29]

Example 1

This example uses the following ABG results: pH 7.39; PaCO₂ 51 mm Hg; PaO₂ 59 mm Hg; HCO₃ 30 mEq/L; and SaO₂ 90%, on room air.[28]

- The pH is within the normal range, so 7.40 is used as the cutoff; in this case, the pH is less than 7.40, indicating acidosis.
- The elevated PaCO₂ indicates respiratory acidosis, and the elevated HCO₃ indicates a metabolic alkalosis.
- The value consistent with the pH is PaCO₂. Therefore, this case is a primary respiratory acidosis. The acid-base component that is inconsistent with the pH is the elevated HCO₃, indicating a compensatory metabolic alkalosis. Because metabolic compensation takes days to be effective, this finding suggests a non-acute primary disorder.
- Finally, the decreased PaO₂ indicates an abnormality with oxygenation. Obtaining a detailed history and performing a thorough physical examination help determine the severity and urgency of any required interventions.

Example 2

This example uses the following ABG results: pH 7.45; PaCO₂ 32 mm Hg; PaO₂ 138 mm Hg; HCO₃ 23 mEq/L; base deficit 1 mEq/L; and SaO₂ 92%, on room air.[28]

- The pH is within the normal range. With 7.40 as the cutoff, the pH is greater than 7.40, indicating alkalemia.
- The decreased PaCO₂ indicates a respiratory alkalosis. The HCO₃ is on the low end of normal.
- The value consistent with the pH is PaCO₂. Therefore, this case is a primary respiratory alkalosis. The HCO₃ is within the normal range and thus is not inconsistent with the pH; this finding indicates a lack of compensation.
- Finally, the PaO₂ is within the normal range, so oxygenation is unaffected.

When evaluating a patient's acid-base status, assessing for an electrolyte imbalance or an anion gap is important for synthesizing the information.[31] For example, a patient who presents with DKA eliminates ketones and closes the anion gap, but the metabolic acidosis persists due to hyperchloremia.

Clinical Significance

Arterial blood gas monitoring is the standard for assessing a patient's oxygenation, ventilation, and acid-base status. ABGs are frequently performed to evaluate oxygenation in the context of severe sepsis, acute respiratory failure, and ARDS in the intensive care unit and emergency room settings. Although ABG monitoring has largely been replaced by noninvasive monitoring, ABGs remain useful for confirming and calibrating noninvasive monitoring techniques.[1]

Calculating an alveolar-arterial oxygen gradient can help narrow down the cause of a patient's hypoxemia.[25] For example, the presence or absence of an A-a gradient can help determine whether the abnormality in oxygenation is potentially due to hypoventilation, a shunt, V/Q mismatch, or impaired diffusion. The equation for the expected A-a gradient assumes the patient is breathing room air; therefore, it is less accurate at higher fractions of inspired oxygen (FiO₂). Determining the intrapulmonary shunt fraction (ie, the fraction of cardiac output flowing through pulmonary units that do not contribute to gas exchange) is the best estimate of oxygenation status. The shunt fraction is traditionally calculated while the patient is breathing an FiO₂ of 1.0; when calculated at lower FiO₂ values, the term *venous admixture* is more appropriate.[1]

For simplicity, oxygenation is more commonly assessed using the ratio of PaO₂ to FiO₂, commonly referred to as the P/F ratio. However, using the P/F ratio in assessing oxygenation carries its own limitations, as the discrepancy between venous admixture and the P/F ratio at a given shunt fraction depends on the delivered FiO₂. Researchers use the P/F ratio to categorize disease severity in ARDS.[32]

Another parameter commonly used in intensive care units to assess oxygenation is the oxygenation index. This index is considered a better indicator of lung injury, particularly in neonatal and pediatric populations, than the P/F ratio. This index also includes the level of invasive ventilatory support required to maintain oxygenation.[33] The oxygenation index is the product of the mean airway pressure (Paw) in cm H₂O, as measured by the ventilator, and the FiO₂ percentage divided by the PaO₂. The oxygenation index is commonly used to guide management, including initiating inhaled nitric oxide, administering surfactant, and determining the potential need for extracorporeal membrane oxygenation.[34]

A normal PaO₂ value does not rule out respiratory failure, particularly in the presence of supplemental oxygen. PaCO₂ is a more sensitive marker of ventilatory failure than PaO₂, particularly in the presence of supplemental oxygen, as it is closely related to the depth and rate of breathing.[27] The PaCO₂ reflects pulmonary ventilation and cellular CO₂ production. Calculating the pulmonary dead space is a good indicator of overall lung function. Pulmonary dead space is the difference between the PaCO₂ and mixed expired PCO₂ (physiological dead space) or

the end-tidal PCO₂ divided by the PaCO₂. Pulmonary dead space increases when ventilation in pulmonary units rises relative to perfusion, while shunting also increases. Hence, pulmonary dead space is an excellent bedside indicator of lung function and one of the best prognostic factors in patients with ARDS.[1] The pulmonary dead space fraction may also help diagnose other conditions, such as pulmonary embolism.[35]

Acid-base balance can be affected by the aforementioned respiratory system abnormalities. For instance, acute respiratory acidosis and alkalemia result in acidemia and alkalemia, respectively. Additionally, hypoxemic hypoxia leads to anaerobic metabolism, which causes metabolic acidosis that results in acidemia. Metabolic system abnormalities also affect acid-base balance, as acute metabolic acidosis and alkalosis similarly result in acidemia and alkalemia.[25] Patients with DKA, septic shock, renal failure, drug or toxin ingestion, and gastrointestinal or renal HCO₃ loss exhibit metabolic acidosis.[28] Conditions such as kidney disease, electrolyte imbalances, prolonged vomiting, hypovolemia, diuretic use, and hypokalemia cause metabolic alkalosis.[36]

Quality Control and Lab Safety

Healthcare providers can analyze an ABG and electrolytes, often called a shock panel, using point-of-care testing (POCT). Appropriate calibration or standardization of these machines ensures accurate and precise readings for clinical decisions. Clinicians should refer to the user manuals to ensure appropriate device calibration when discussing with the clinical laboratory team.[37]

Elements of good quality assurance of blood gas and pH measurements include the following:

- Proper maintenance of the instrument
- Use of control materials
- Verification of electrode linearity
- Checking barometer accuracy
- Accurate measurement of temperature [38]

External quality assurance (proficiency testing) mandated by federal law in the United States (Clinical Laboratory Improvement Amendments 1988) assumed importance for quality control of blood gas analysis.[39] These rules became effective in January 1991 and set criteria for satisfactory interlaboratory performance, which are as follows:

- pH: Target value ± 0.04
- PO₂: Target value ± 3 SD
- PCO₂: Target value $\pm 8\%$ or ± 5 mm Hg, whichever is greater [40]

The importance of proficiency testing and the penalties for failure place strong incentives on consistent performance of internal control measures and effective response to quality control failures.[41] At the same time, pressure to control healthcare costs has raised questions about how frequently interlaboratory performance should be monitored and which concentrations of control materials are necessary. According to Clinical Laboratory Improvement Amendments 1988, the answer is 1 concentration of control every 8 hours, with the entire range of control concentrations covered every 24 hours.[22] In many laboratories, however, the practical answer is to run 3 concentrations of control for pH, PO₂, and PCO₂ on every instrument in use at least once per shift and always upon completion of maintenance and troubleshooting procedures.

Newer analyzers, particularly smaller satellite laboratory and point-of-care instruments, frequently have automatic quality control (QC) features or use electronic QC.[42] Auto QC uses internally stored QC materials, which are automatically analyzed at designated intervals to meet regulatory requirements. Electronic QC, most commonly

incorporated into devices with disposable electrode cartridges, uses cartridge-based systems to verify instrument electronic performance specifications.[43]

From a laboratory safety perspective, ABG analysis in both central laboratories and POCT sites requires strict adherence to biosafety and infection control practices to minimize the risk of exposure to bloodborne pathogens and pre-analytical errors. All ABG specimens are considered potentially infectious and must be handled using appropriate personal protective equipment, including gloves, lab coats, and eye protection, along with strict hand hygiene before and after sample handling. In the central laboratory, safety is ensured through proper specimen transport in sealed, leak-proof, anaerobically maintained syringes, immediate analysis to reduce biohazard exposure time, and safe disposal of sharps in designated containers to prevent needle-stick injuries.

In POCT settings such as ICUs and emergency departments, additional safety emphasis is placed on operator competency, device disinfection between patients, and strict adherence to cartridge-based system handling to prevent cross-contamination. Both settings require meticulous attention to prevent air contamination, sample leakage, and improper handling of heparinized syringes, as these factors not only compromise analytical accuracy but also increase occupational risk. Implementation of standardized operating procedures, regular training, and compliance with institutional biosafety and quality standards help ensure the safe practice of ABG testing across all hospital environments.

Enhancing Healthcare Team Outcomes

An ABG can be used to assess a patient's ventilatory, acid-base, and oxygenation status. Additionally, blood gas analysis is recommended to assess a patient's response to therapeutic interventions and to monitor the severity and progression of documented cardiopulmonary disease processes.[44] Despite its clinical value, erroneous or discrepant values represent a potential drawback of blood gas analysis, so eliminating potential sources of error is paramount. [27] Therefore, attention to detail in the sampling technique and processing is essential.

Rigorous quality control of automated blood gas analyzers is also necessary to ensure accurate results. However, advances in machine performance and quality assurance have now made most errors in POCT ABG analysis attributable to clinical providers. Several pre-analytic steps must be followed to obtain a valid, interpretable ABG. [27] In most hospital settings, ABG analysis involves multiple healthcare providers (eg, clinicians, respiratory therapists, and nurses). Hence, interprofessional coordination, cooperation, and communication are vitally important.

The American Association for Respiratory Care has published Clinical Care Guidelines for Blood Gas Analysis and Hemoximetry that provide current best practices for sampling, handling, and analyzing ABGs.[44] Notable sources of erroneous values during blood draws include abnormal or misstated FiO_2 , barometric pressures, or temperatures. Temperature is a significant variable that can lead to discrepancies between PaO_2 and SaO_2 , as can acid-base disturbances. Several physiological and clinical conditions, including hyperleukocytosis and dyshemoglobinemias, may also result in discrepancies between PaO_2 and SaO_2 . Sample dilution can be an additional source of error, with liquid heparin and saline as potential contributors.[45]

The mode of sample transport is also significant, as discrepant values can result from air contamination during pneumatic tube transport compared with manual transport of the specimen, especially in the presence of inadvertent air bubbles.[45] Therefore, procuring samples using suitable syringes filled with adequate amounts of blood, free of air bubbles, maintained at the correct temperatures, and transporting them appropriately and promptly for rapid analysis can minimize erroneous values.

Review Questions

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References

1. Gattinoni L, Pesenti A, Matthay M. Understanding blood gas analysis. *Intensive Care Med.* 2018 Jan;44(1):91-93. [PubMed: 28497267]
2. Boemke W, Krebs MO, Rossaint R. [Blood gas analysis]. *Anaesthesist.* 2004 May;53(5):471-92; quiz 493-4. [PubMed: 15222335]
3. Ziegenfuß T, Zander R. Understanding blood gas analysis. *Intensive Care Med.* 2019 Nov;45(11):1684-1685. [PubMed: 31392344]
4. Jubran A. Pulse oximetry. *Crit Care.* 2015 Jul 16;19(1):272. [PMC free article: PMC4504215] [PubMed: 26179876]
5. Kumar V, Karon BS. Comparison of measured and calculated bicarbonate values. *Clin Chem.* 2008 Sep;54(9):1586-7. [PubMed: 18755915]
6. Matoušovic K, Havlín J, Schück O. [Clinical evaluation of acid-base status: Henderson-Hasselbalch, or Stewart-Fencel approach?]. *Cas Lek Cesk.* 2016 Winter;155(7):365-369. [PubMed: 27990831]
7. Kim Y, Massie L, Murata GH, Tzamaloukas AH. Discrepancy between Measured Serum Total Carbon Dioxide Content and Bicarbonate Concentration Calculated from Arterial Blood Gases. *Cureus.* 2015 Dec 07;7(12):e398. [PMC free article: PMC4725444] [PubMed: 26824002]
8. Hopkins SR. Ventilation/Perfusion Relationships and Gas Exchange: Measurement Approaches. *Compr Physiol.* 2020 Jul 08;10(3):1155-1205. [PMC free article: PMC8274320] [PubMed: 32941684]
9. Dev SP, Hillmer MD, Ferri M. Videos in clinical medicine. Arterial puncture for blood gas analysis. *N Engl J Med.* 2011 Feb 03;364(5):e7. [PubMed: 21288091]
10. Malatesha G, Singh NK, Bharija A, Rehani B, Goel A. Comparison of arterial and venous pH, bicarbonate, PCO₂ and PO₂ in initial emergency department assessment. *Emerg Med J.* 2007 Aug;24(8):569-71. [PMC free article: PMC2660085] [PubMed: 17652681]
11. Magnet FS, Majorski DS, Callegari J, Schwarz SB, Schmoor C, Windisch W, Storre JH. Capillary PO₂ does not adequately reflect arterial PO₂ in hypoxemic COPD patients. *Int J Chron Obstruct Pulmon Dis.* 2017;12:2647-2653. [PMC free article: PMC5593412] [PubMed: 28919732]
12. Hill S, Moore S. Arterial blood gas sampling: using a safety and pre-heparinised syringe. *Br J Nurs.* 2018 Jul 26;27(14):S20-S26. [PubMed: 30048173]
13. Higgins C. The use of heparin in preparing samples for blood-gas analysis. *MLO Med Lab Obs.* 2007 Oct;39(10):16-8, 20; quiz 22-3. [PubMed: 18018679]
14. Jiang HX. [The effect of dilution and heparin on the blood gas analysis]. *Zhonghua Jie He He Hu Xi Za Zhi.* 1992 Aug;15(4):225-7, 255-6. [PubMed: 1307518]
15. Sood P, Paul G, Puri S. Interpretation of arterial blood gas. *Indian J Crit Care Med.* 2010 Apr;14(2):57-64. [PMC free article: PMC2936733] [PubMed: 20859488]
16. Dukić L, Kopčinović LM, Dorotić A, Baršić I. Blood gas testing and related measurements: National recommendations on behalf of the Croatian Society of Medical Biochemistry and Laboratory Medicine. *Biochem Med (Zagreb).* 2016 Oct 15;26(3):318-336. [PMC free article: PMC5082214] [PubMed: 27812301]
17. Knowles TP, Mullin RA, Hunter JA, Douce FH. Effects of syringe material, sample storage time, and temperature on blood gases and oxygen saturation in arterialized human blood samples. *Respir Care.* 2006 Jul;51(7):732-6. [PubMed: 16800906]
18. Çuhadar S, Özkanay-Yörük H, Köseoğlu M, Katırcıoğlu K. Detection of preanalytical errors in arterial blood gas analysis. *Biochem Med (Zagreb).* 2022 Jun 15;32(2):020708. [PMC free article: PMC9195608] [PubMed: 35799987]
19. Lima-Oliveira G, Lippi G, Salvagno GL, Montagnana M, Picheth G, Guidi GC. Different manufacturers of syringes: a new source of variability in blood gas, acid-base balance and related laboratory test? *Clin Biochem.* 2012 Jun;45(9):683-7. [PubMed: 22440459]
- 20.

- Kirubakaran C, Gnananayagam JE, Sundaravalli EK. Comparison of blood gas values in arterial and venous blood. *Indian J Pediatr*. 2003 Oct;70(10):781-5. [PubMed: 14649471]
21. Beutel BG, Worley C, Zisquit J, Nedeff N. StatPearls [Internet]. StatPearls Publishing; Treasure Island (FL): Oct 14, 2024. Allen Test. [PubMed: 29939593]
 22. Rodríguez-Villar S, Poza-Hernández P, Freigang S, Zubizarreta-Ormazabal I, Paz-Martín D, Holl E, Pérez-Pardo OC, Tovar-Doncel MS, Wissa SM, Cimadevilla-Calvo B, Tejón-Pérez G, Moreno-Fernández I, Escario-Méndez A, Arévalo-Serrano J, Valentín A, Do-Vale BM, Fletcher HM, Lorenzo-Fernández JM. Automatic real-time analysis and interpretation of arterial blood gas sample for Point-of-care testing: Clinical validation. *PLoS One*. 2021;16(3):e0248264. [PMC free article: PMC7946183] [PubMed: 33690724]
 23. José RJ, Preller J. Near-patient testing of potassium levels using arterial blood gas analysers: can we trust these results? *Emerg Med J*. 2008 Aug;25(8):510-3. [PubMed: 18660404]
 24. Adekola OO, Soriyan OO, Meka I, Akanmu ON, Olanipekun S, Oshodi TA. The incidence of electrolyte and acid-base abnormalities in critically ill patients using point of care testing (i-STAT portable analyser). *Nig Q J Hosp Med*. 2012 Apr-Jun;22(2):103-8. [PubMed: 23175907]
 25. Yee J, Frinak S, Mohiuddin N, Uduman J. Fundamentals of Arterial Blood Gas Interpretation. *Kidney360*. 2022 Aug 25;3(8):1458-1466. [PMC free article: PMC9416819] [PubMed: 36176645]
 26. Arbiol-Roca A, Imperiali CE, Dot-Bach D, Valero-Politi J, Dastis-Arias M. Stability of pH, Blood Gas Partial Pressure, Hemoglobin Oxygen Saturation Fraction, and Lactate Concentration. *Ann Lab Med*. 2020 Nov;40(6):448-456. [PMC free article: PMC7295962] [PubMed: 32539300]
 27. Cowley NJ, Owen A, Bion JF. Interpreting arterial blood gas results. *BMJ*. 2013 Jan 16;346:f16. [PubMed: 23325867]
 28. Larkin BG, Zimmanck RJ. Interpreting Arterial Blood Gases Successfully. *AORN J*. 2015 Oct;102(4):343-54; quiz 355-7. [PubMed: 26411819]
 29. Rogers KM, McCutcheon K. Four steps to interpreting arterial blood gases. *J Perioper Pract*. 2015 Mar;25(3):46-52. [PubMed: 26016282]
 30. Romanski SO. Interpreting ABGs in four easy steps (continuing education credit). *Nursing*. 1986 Sep;16(9):58-64. [PubMed: 3638536]
 31. Seifter JL. Integration of acid-base and electrolyte disorders. *N Engl J Med*. 2014 Nov 06;371(19):1821-31. [PubMed: 25372090]
 32. ARDS Definition Task Force. Ranieri VM, Rubenfeld GD, Thompson BT, Ferguson ND, Caldwell E, Fan E, Camporota L, Slutsky AS. Acute respiratory distress syndrome: the Berlin Definition. *JAMA*. 2012 Jun 20;307(23):2526-33. [PubMed: 22797452]
 33. Pediatric Acute Lung Injury Consensus Conference Group. Pediatric acute respiratory distress syndrome: consensus recommendations from the Pediatric Acute Lung Injury Consensus Conference. *Pediatr Crit Care Med*. 2015 Jun;16(5):428-39. [PMC free article: PMC5253180] [PubMed: 25647235]
 34. Rawat M, Chandrasekharan PK, Williams A, Gugino S, Koenigsnecht C, Swartz D, Ma CX, Mathew B, Nair J, Lakshminrusimha S. Oxygen saturation index and severity of hypoxic respiratory failure. *Neonatology*. 2015;107(3):161-6. [PMC free article: PMC4405613] [PubMed: 25592054]
 35. Kurt OK, Alpar S, Sipit T, Guven SF, Erturk H, Demirel MK, Korkmaz M, Hayran M, Kurt B. The diagnostic role of capnography in pulmonary embolism. *Am J Emerg Med*. 2010 May;28(4):460-5. [PubMed: 20466226]
 36. Wagner PD. The physiological basis of pulmonary gas exchange: implications for clinical interpretation of arterial blood gases. *Eur Respir J*. 2015 Jan;45(1):227-43. [PubMed: 25323225]
 37. Nichols JH. Quality in point-of-care testing. *Expert Rev Mol Diagn*. 2003 Sep;3(5):563-72. [PubMed: 14510177]
 38. O'Shaughnessy P, Emancipater K, Hsu C. An assessment of quality control testing in an emergency department (ED) maintained arterial blood gas analyzer. *Acad Emerg Med*. 2000 Oct;7(10):1168. [PubMed: 11015269]
 39. Ehrmeyer SS, Laessig RH. Has compliance with CLIA requirements really improved quality in US clinical laboratories? *Clin Chim Acta*. 2004 Aug 02;346(1):37-43. [PubMed: 15234634]

40. Final CLIA regulations. Clinical Laboratory Improvement Amendments. Health Devices. 1992 Nov;21(11):420-5. [PubMed: 1428896]
41. Dunn SG. Point-of-care testing: the impact of CLIA regulations on your testing program. J Med Pract Manage. 2007 May-Jun;22(6):341-5. [PubMed: 17612308]
42. Tucker AM, Johnson TN. Acid-base disorders: A primer for clinicians. Nutr Clin Pract. 2022 Oct;37(5):980-989. [PubMed: 35752932]
43. Dukić L, Simundić AM. Institutional practices and policies in acid-base testing: a self reported Croatian survey study on behalf of the Croatian society of medical biochemistry and laboratory medicine Working Group for acid-base balance. Biochem Med (Zagreb). 2014;24(2):281-92. [PMC free article: PMC4083580] [PubMed: 24969922]
44. Davis MD, Walsh BK, Sittig SE, Restrepo RD. AARC clinical practice guideline: blood gas analysis and hemoximetry: 2013. Respir Care. 2013 Oct;58(10):1694-703. [PubMed: 23901131]
45. Albert TJ, Swenson ER. Circumstances When Arterial Blood Gas Analysis Can Lead Us Astray. Respir Care. 2016 Jan;61(1):119-21. [PubMed: 26682966]

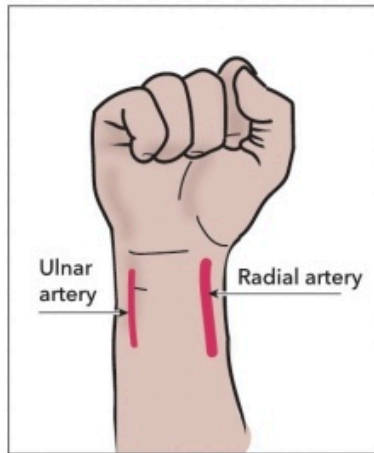
Disclosure: Danny Castro declares no relevant financial relationships with ineligible companies.

Disclosure: Muhammad Zubair declares no relevant financial relationships with ineligible companies.

Figures

The Modified Allen Test

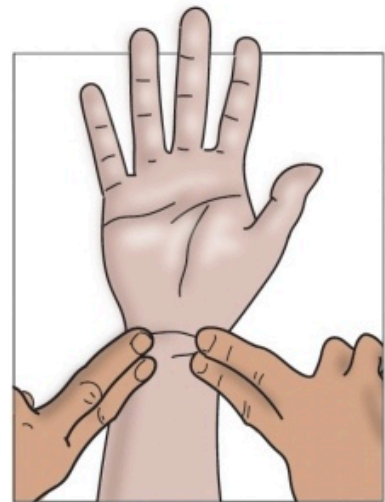
This test is used to check the overall blood supply to the hand.



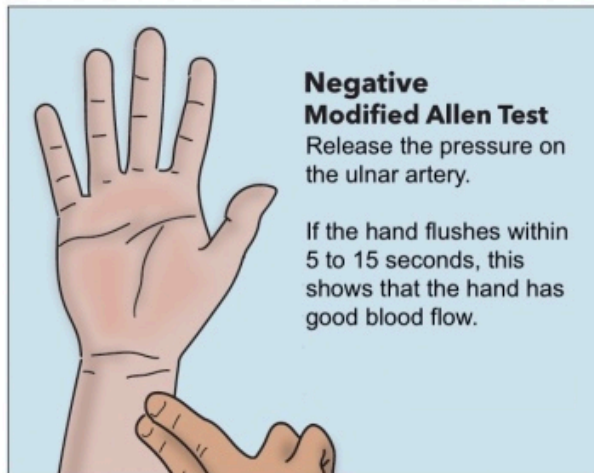
Locate the ulnar and radial arteries. Have the patient make a tight fist for about 30 seconds.



To obstruct blood flow, press down on the ulnar artery with 2 fingers. At the same time, press down on the radial artery.



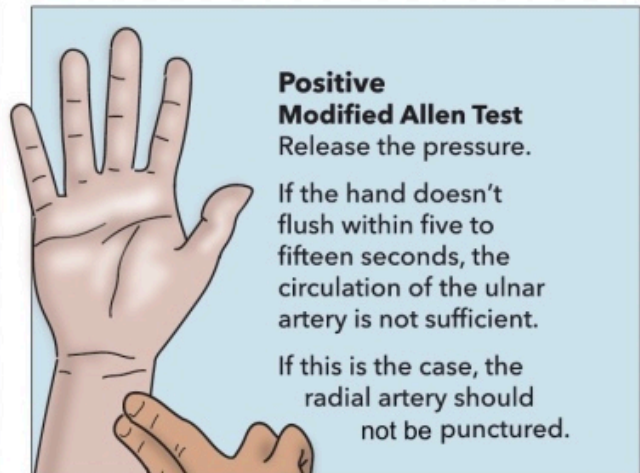
Tell the patient to unclench; their palm should blanch. If it does not, you are not applying enough pressure—start again.



Negative Modified Allen Test

Release the pressure on the ulnar artery.

If the hand flushes within 5 to 15 seconds, this shows that the hand has good blood flow.



Positive Modified Allen Test

Release the pressure.

If the hand doesn't flush within five to fifteen seconds, the circulation of the ulnar artery is not sufficient.

If this is the case, the radial artery should not be punctured.

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Modified Allen Test. This test is used to check the overall blood supply to the hand. Illustrated by K Humphreys

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